

Lecture - 14 - Expectation-Maximization Algorithms.

K-Means Clustering

Data: $\{x^{(1)}, x^{(2)}, \dots, x^{(m)}\}$

1) Initialize cluster centroids

$\mu_k \leftarrow \mathbb{R}^n$ randomly.

2) Repeat until convergence

(a) Set $c^{(i)} = \underset{j}{\operatorname{argmin}} \|x^{(i)} - \mu_j\|^2$ (color the points)

(b) For $j=1, \dots, k$

$$\mu_j = \frac{\sum_{i=1}^m \mathbb{1}\{c^{(i)} = j\} x^{(i)}}{\sum_{i=1}^m \mathbb{1}\{c^{(i)} = j\}}$$

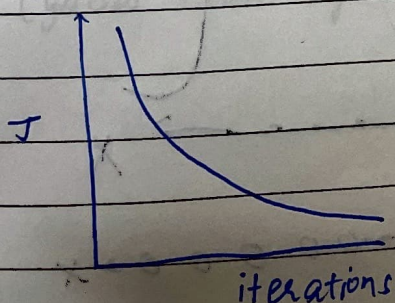
(moves the cluster to centroid)

Cost Function

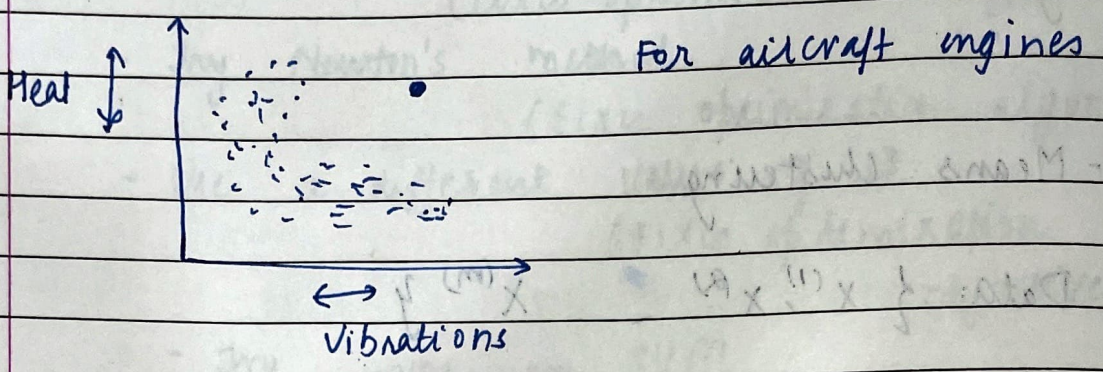
$$J(c, \mu) = \sum_{i=1}^m \|x^{(i)} - \mu_{c^{(i)}}\|^2$$

centroids

assignments



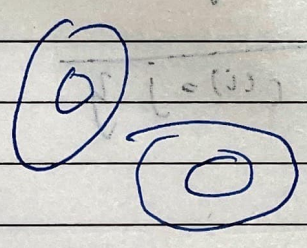
Density Estimation



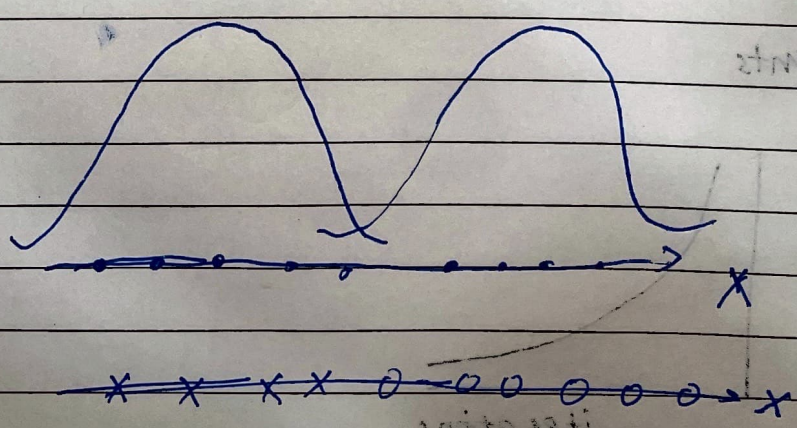
For anomaly detection, Model $p(x)$ if $p(x) < \epsilon$

prob ability,
then it can be
called as
anomaly.

There is generally no L shaped distribution, but these can be interpreted as 2 gaussians.



1-D Example $x \in \mathbb{R}$



Mixture of Gaussian Models

Suppose there's a latent (hidden/unobserved) random variables z_i , and $x^{(i)}, z^{(i)}$ are joint distributed

$$P(x^{(i)}, z^{(i)}) = P(x^{(i)} | z^{(i)}) P(z^{(i)}) \quad z \in \{1, \dots, k\}$$

where $z^{(i)} \sim \text{Multinomial}(\phi)$

$$x^{(i)} | z^{(i)} = j \sim \mathcal{N}(\mu_j, \Sigma_j) \quad \text{sigma}$$

If we know the $z^{(i)}$'s, we can use the MLE

$$L(\phi, \mu, \Sigma) = \sum_{i=1}^m \log(P(x^{(i)}, z^{(i)}; \phi, \mu, \Sigma))$$

$$\phi_j = \frac{1}{m} \sum_{i=1}^m \mathbb{1}\{z^{(i)} = j\} \quad \text{Take derivative \& equate to 0.}$$

$$\mu_j = \frac{\sum_{i=1}^m \mathbb{1}\{z^{(i)} = j\} x^{(i)}}{\sum_{i=1}^m \mathbb{1}\{z^{(i)} = j\}}$$

~~E-step~~
 EM Algorithm

E-step (Guess value of $z^{(i)}$'s)

Set $w_j^{(i)} = p(z^{(i)} = j | x^{(i)}; \phi, \mu, \Sigma)$

$$= \frac{p(x^{(i)} | z^{(i)} = j) p(z^{(i)} = j)}{\sum_{l=1}^K p(x^{(i)} | z^{(i)} = l) \cdot p(z^{(i)} = l)}$$

comes from

Gaussian

density

$$\frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x^{(i)} - \mu_j)^T \Sigma^{-1} (x^{(i)} - \mu_j)\right)$$

Also, $p(z^{(i)} = j) = \phi_j$; $z \sim \text{multinomial}(\phi)$

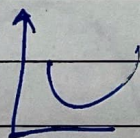
m-step

$$\phi_j = \frac{1}{m} \sum_{i=1}^m w_j^{(i)} \quad w_j^{(i)} = \mathbb{E}[\mathbb{1}_{\{z^{(i)} = j\}}]$$

$$\mu_j = \sum_{i=1}^m w_j^{(i)} x^{(i)} \quad \text{indicator function.}$$

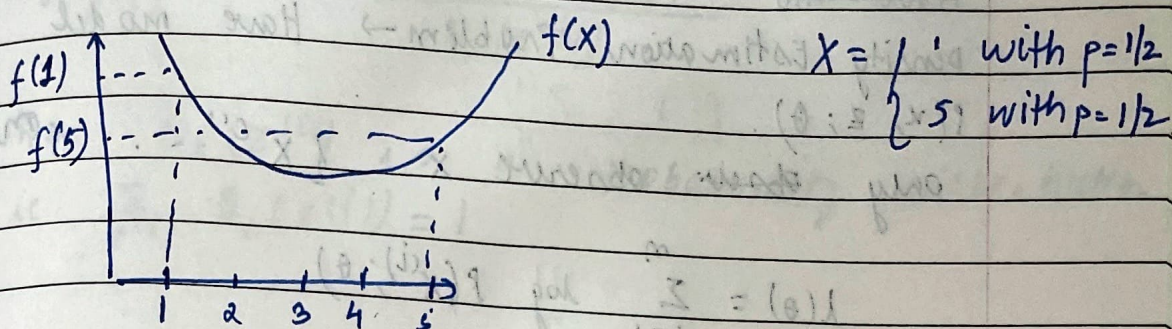
Jensen's Inequality

Let f be a convex function, such as $f''(x) > 0$



Let x be a random variable, then $f(\mathbb{E}x) \leq \mathbb{E}[f(x)]$

Example



$$f(3) = f(EX)$$

$$E[f(X)] = \frac{1}{2} f(1) + \frac{1}{2} f(5)$$

So by Jensen's equality:

$$f(EX) \leq E[f(X)]$$

Further, if $f''(x) > 0$ (f is strictly convex) then

$$E[f(X)] = f(EX) \Leftrightarrow X \text{ is a constant.}$$

Jensen's inequality for concave functions. So for concave function, $f''(x) < 0$.

Let X be a random variable, then $f(EX) \geq E[f(X)]$

Further, if $f''(x) < 0$ (f is strictly concave), then

$$E[f(X)] = f(EX) \Leftrightarrow X \text{ is a constant.}$$

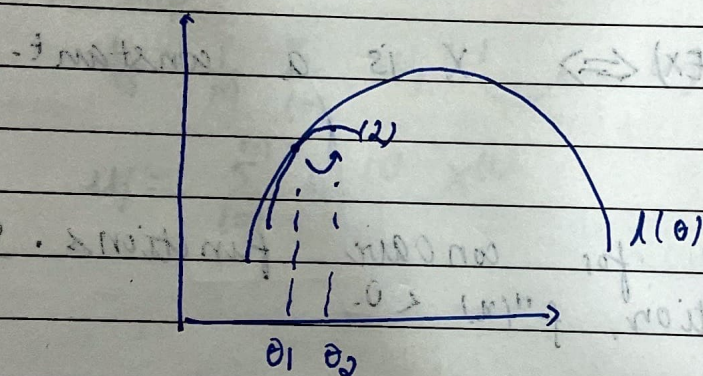
Have ~~no~~ Density Estimation Problem $\rightarrow$ Have model for $P(x, z; \theta)$.

only observe $x \in \{x^{(1)}, \dots, x^{(m)}\}$

$$l(\theta) = \sum_{i=1}^m \log P(x^{(i)}; \theta)$$

$$= \sum_{i=1}^m \log \sum_{z^{(i)}} P(x^{(i)}, z^{(i)}; \theta)$$

The ~~Density Estimation Problem~~ is that we want value of θ which maximizes $\log l(\theta)$



In the EM algorithm, the E step is to draw curve (2), and the m-step ~~to~~ takes the curve (2), and finds the maximum θ_2 .

$$\max_{\theta} \sum_i \log (P(x^{(i)}; \theta))$$

$$= \sum_i \log \sum_{z^{(i)}} P(x^{(i)}, z^{(i)}; \theta)$$

$$\sum_i \log \sum_{z^{(i)}} Q_i(z^{(i)}) \frac{P(x^{(i)}, z^{(i)}; \theta)}{Q_i(z^{(i)})}$$

where $Q_i(z^{(i)})$ is a probability distribution
i.e. $\sum_{z^{(i)}} Q_i(z^{(i)}) = 1$

$$\sum_i \log E_{z^{(i)} \sim Q_i} \left[\frac{P(x^{(i)}, z^{(i)}; \theta)}{Q_i(z^{(i)})} \right]$$



applying concave form of Jensen's inequality

$$\geq \sum_i E_{z^{(i)} \sim Q_i} \left[\log \frac{P(x^{(i)}, z^{(i)}; \theta)}{Q_i(z^{(i)})} \right]$$

$$= \sum_i \sum_{z^{(i)}} Q_i(z^{(i)}) \log \frac{P(x^{(i)}, z^{(i)}; \theta)}{Q_i(z^{(i)})}$$

side note $\rightarrow z \in \{1, \dots, 10\} \rightarrow Q_i(z)$

$$E[g(z)] = \sum_z p(z) \cdot g(z)$$

$$E[z] = \sum_z p(z) \cdot z$$

on a given iteration of EM with parameter θ , we want.

$$\log E_{z^{(i)} \sim Q_i} \left[\frac{P(x^{(i)}, z^{(i)}; \theta)}{Q_i(z^{(i)})} \right] = E_{z^{(i)} \sim Q_i} \left[\log \frac{P(x^{(i)}, z^{(i)}; \theta)}{Q_i(z^{(i)})} \right]$$

$$\log \left[\frac{P(x^{(i)}, z^{(i)}; \theta)}{Q_i(z^{(i)})} \right]$$

For this to hold, we need

$$\frac{P(x^{(i)}, z^{(i)})}{Q_i(z^{(i)})} = \text{constant}$$

$$\text{Set } Q_i(z^{(i)}) \propto P(x^{(i)}, z^{(i)}, \theta)$$

$$\sum_{z^{(i)}} Q_i(z^{(i)}) = 1$$

$$Q_i(z^{(i)}) = \frac{P(x^{(i)}, z^{(i)}; \theta)}{\sum_{z^{(i)}} P(x^{(i)}, z^{(i)}; \theta)}$$

From
lecture
notes

$$= P(z^{(i)} | x^{(i)}, \theta)$$

E-step

$$\text{Set } Q_i(z^{(i)}) = P(z^{(i)} | x^{(i)}, \theta)$$

m-step:

$$\theta := \arg \max_{\theta} \sum_i \sum_{z^{(i)}} Q_i(z^{(i)}) \log \frac{P(x^{(i)}, z^{(i)}, \theta)}{Q_i(z^{(i)})}$$